

Section : Physics

Q.1 A circular loop of area 0.02m^2 carrying a current of 5 A is held with its plane normal to a magnetic field of 0.2 T.

The torque acting on the loop is

1. 0.002 N m
2. 0.02 N m
3. 0.2 N m
4. 0

Options 1. 1

2. 2
3. 3
4. 4

Q.2 A battery of emf 4 V and internal resistance $1\ \Omega$ is connected across a wire of resistance $9.0\ \Omega$.

The number of electrons passing through the cross-section of the wire in 4 s is:

1. 10^{19}
2. 10^{20}
3. 10^{18}
4. 2.5×10^{18}

Options 1. 1

2. 2
3. 3
4. 4

Q.3 A current change of 6 A in one coil causes the flux in the second coil of 2000 turns

to change by 12×10^{-4} Wb per turn. The mutual inductance of the pair of coils would be:

1. 4.0 H
2. 0.4 H
3. 2.0 H
4. 0.2 H

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Two long and parallel straight wires, each carrying current I in the same direction, are separated by a distance of 40 cm.

If the force of attraction per metre between them is 2×10^{-6} N/m, the value of I will be:

(Given: $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m}\cdot\text{A}^{-1}$)

1. 0.56 A
2. 0.63 A
3. 2 A
4. 4 A

Options 1. 1

2. 2
3. 3
4. 4

Q.5 An electron (mass 9×10^{-31} kg and charge 1.6×10^{-19} C) moving with a speed of 5.0×10^7 m/s

enters a region of uniform magnetic field of strength 0.04 T which is perpendicular to its velocity. The radius of the circular path it follows is:

1. 7.11×10^{-5} m
2. 8.05×10^{-5} m
3. 7.03×10^{-3} m
4. 8.32×10^{-3} m

Options 1. 1

2. 2
3. 3
4. 4

Q.6 The kinetic energy of an electron in the 2nd excited state in a hydrogen atom, if its ground state energy is -13.6 eV, will be:

1. 13.6 eV
2. 3.4 eV
3. 1.51 eV
4. 0.85 eV

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Arrange the given associated wavelengths (λ_1 , λ_2 , λ_3 and λ_4) of radiations with the part of the electromagnetic spectrum in decreasing order of magnitude:

- (A) λ_1 used to detect fracture in bones
- (B) λ_2 used to kill germs in water purifiers
- (C) λ_3 used to treat muscular strain
- (D) λ_4 used to transmit voice communication

Choose the correct answer from the options given below:

- 1. (D), (C), (B), (A)
- 2. (A), (B), (D), (C)
- 3. (B), (A), (D), (C)
- 4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.8 Which of the following statements is INCORRECT?

- 1. A moving charge produces a magnetic field which in turn exerts a force on another moving charge.
- 2. An electric charge always experiences a force in an electric field, whether the charge is stationary or in motion.
- 3. A charge moving parallel or antiparallel to the direction of the magnetic field experiences a magnetic Lorentz force.
- 4. A stationary charge does not produce any magnetic field to interact with an external magnetic field.

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.9 The radius of a wire having resistance R is made half and its length reduced to one - fourth of its original length. The new resistance of the wire will be

- 1. $R/2$
- 2. R
- 3. $4R$
- 4. $2R$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.10 Read the following statements related to telescopes carefully:

- (A) The main considerations of an astronomical telescope are its light gathering power and its resolution.
- (B) the ability to observe two objects (very nearly in the same direction) distinctly depends on the diameter of the objective
- (C) large-sized lenses which form images that are free from any kind of chromatic aberration and distortions can be made easily
- (D) Telescopes with mirror objectives are free from chromatic aberration

Choose the correct answer from the options given below:

- 1. (A), (B) and (C) only
- 2. (A), (B) and (D) only
- 3. (B), (C) and (D) only
- 4. (A), (B), (C) and (D)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.11 The number of electric lines of force originating from a charge of 1 C is:

- 1. 11.29×10^{16}
- 2. 1.129×10^8
- 3. 1.129×10^{11}
- 4. 1.129×10^{15}

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.12 Which of the following is not the use of a shunt?

- 1. To prevent a galvanometer from being damaged due to a large current
- 2. To increase the range of voltmeter
- 3. To convert a galvanometer into an ammeter
- 4. To increase the range of an ammeter

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.13 A parallel plate air capacitor consists of two circular plates of diameter 4 cm. At what distance should the plates be held so as to have the same capacitance as that of a sphere of diameter 16 cm?

1. 6.0 mm
2. 4.0 mm
3. 2.5 mm
4. 1.25 mm

Options 1. 1
2. 2
3. 3
4. 4

Q.14 A coil C_1 is connected to a galvanometer G. Second coil C_2 is connected to a battery through a tapping key. The steady current in the coil C_2 produces a steady magnetic field. As coil C_2 is moved towards the coil C_1 , the galvanometer shows a deflection. To obtain a large deflection, one or more of the following steps can be taken:

- (A) Use a rod made of soft iron inside the coil C_2
- (B) Connect the coil C_2 to a more powerful battery
- (C) Move the arrangement of coil C_2 rapidly towards the coil C_1
- (D) hold the key pressed continuously

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (C) only
3. (A), (B) and (D) only
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.15 If the length of a wire is doubled while keeping its material and cross-sectional area same, the resistance of the wire

1. becomes double of its initial value
2. becomes half of its initial value
3. becomes four times of its initial value
4. remains same

Options 1. 1
2. 2
3. 3
4. 4

Q.16 Two cells each of emf E and internal resistance r are connected in such a way that their positive terminals are connected to point X and negative terminals are connected to point Y. The potential difference between X and Y is

1. zero
2. $0.5 E$
3. E
4. $2E$

Options 1. 1
2. 2
3. 3
4. 4

Q.17 The electric potential at the centre of a square of side $\sqrt{2}$ m having charges $90 \mu\text{C}$, $-60 \mu\text{C}$, $30 \mu\text{C}$ and $-40 \mu\text{C}$ at the four corners of the square is:

1. $1.8 \times 10^5 \text{ V}$
2. $1.8 \times 10^4 \text{ V}$
3. $2.2 \times 10^5 \text{ V}$
4. $1.2 \times 10^5 \text{ V}$

Options 1. 1
2. 2
3. 3
4. 4

Q.18 The hydrogen atom is excited from ground state to a state with the principal quantum number $(n) = 4$. The number of spectral lines in the emission spectra will be

1. 6
2. 5
3. 4
4. 3

Options 1. 1
2. 2
3. 3
4. 4

Q.19 A light ray is incident normal to one face of a right-angled isosceles prism and is totally internally reflected at the glass-air interface. If the angle of reflection is $\frac{\pi}{4}$, what can be concluded about the refractive index?

1. $\mu = \sqrt{2}$
2. $\mu < \frac{1}{\sqrt{2}}$
3. $\mu > \sqrt{2}$

4. $\mu < \sqrt{2}$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.20 Light of wavelength 600 nm is incident normally on a slit of width 0.03 cm. The width of the central maximum on the screen placed 1 m away from the slit is

1. 1 mm
2. 2 mm
3. 4 mm
4. 6 mm

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.21 In an AC circuit, if the peak voltage is 20 V, then root mean square (rms) voltage will be

1. 14.14 V
2. 20 V
3. 28.28 V
4. 0V

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.22 The electric field at a distance of 0.20 m from a point charge q is $1.35 \times 10^3 \text{ N}\cdot\text{C}^{-1}$.

The electric potential at the same distance will be:

1. $5.4 \times 10^{-3} \text{ V}$
2. $2.7 \times 10^2 \text{ V}$
3. $6 \times 10^{-9} \text{ V}$
4. $2.7 \times 10^{-4} \text{ V}$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.23 C, Si and Ge have the same lattice structure but C is an insulator while Si and Ge are intrinsic semiconductors because

- (A) ionisation energy will be least for C, followed by Si and highest for Ge
- (B) ionisation energy will be least for Ge, followed by Si and highest for C
- (C) number of free electrons for conduction in Ge and Si are negligibly small but significant for C
- (D) number of free electrons for conduction in Ge and Si are significant but negligibly small for C

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.24 The electric field intensity at a distance of 20 cm from the centre of a small conducting sphere is $1.5 \times 10^3 \text{ N}\cdot\text{C}^{-1}$ and is directed towards the sphere.

The charge on the sphere is:

1. + 3.3 nC
2. - 6.6 nC
3. - 3.3 nC
4. 0

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.25 If the frequency of an alternating current is doubled while keeping the inductance constant, then the inductive reactance

1. is doubled
2. becomes half
3. remains the same
4. becomes four times

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.26 Four identical bulbs are connected in series with a source. These together dissipate a power P. The power dissipated, if

the four bulbs are connected in parallel with the same source, will be

1. $P/4$
2. $P/16$
3. $4P$
4. $16P$

Options

1. 1
2. 2
3. 3
4. 4

Q.27 Two charges, $+20\text{ }\mu\text{C}$ and $+40\text{ }\mu\text{C}$ are placed at a separation of 4 cm. The electric potential due to the pair at the middle point of the line joining the two charges is:

1. 9 MV
2. 18 MV
3. 13.5 MV
4. 27 MV

Options

1. 1
2. 2
3. 3
4. 4

Q.28 A current of 2 A flows through a plane circular coil of radius 3 cm and having 50 turns. The coil has been placed in a uniform magnetic field of 0.6 T. The minimum potential energy of the coil will be

1. 0 J
2. - 0.08 J
3. - 0.17 J
4. 0.17 J

Options

1. 1
2. 2
3. 3
4. 4

Q.29 Some important general properties of the electric field lines are given below:

- (A) Field lines start from positive charges and end at negative charges
- (B) In a charge-free region, electric field lines can be taken to be continuous curves without any breaks
- (C) Electrostatic field lines form closed loops
- (D) Two field lines can never cross each other

Choose the *correct* answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (D) only
3. (B), (C) and (D) only
4. (A), (B), (C) and (D)

Options

1. 1
2. 2
3. 3
4. 4

Q.30 A resistor R and an inductor L are connected in series to a 24 V, 50 Hz supply. A current of 1 A flows through the circuit. If the current differs in phase from the applied voltage by 60° , the value of impedance in the circuit is

1. $24\text{ }\Omega$
2. $12\text{ }\Omega$
3. $17\text{ }\Omega$
4. $6\text{ }\Omega$

Options

1. 1
2. 2
3. 3
4. 4

Q.31 Which of the following statements are correct?

- (A) According to wave theory, energy carried by a wave depends on the amplitude of the wave, not on the speed of wave propagation
- (B) For a given frequency, intensity of light in the photon picture is determined by the number of photons crossing a unit area
- (C) At a large distance from the source, a small portion of the spherical wave can be approximated by a plane wave
- (D) the reflected and refracted light both have different frequencies than the incident frequency

Choose the *correct* answer from the options given below:

1. (A), (B) and (C) only
2. (A) and (C) only
3. (A), (C) and (D) only
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.32 A wire of resistance $20\text{ }\Omega$ is bent to form a complete circle. What will be its resistance between two diametrically opposite points?

1. $40\text{ }\Omega$
2. $20\text{ }\Omega$

3. $10\ \Omega$
4. $5\ \Omega$

Options 1. 1
2. 2
3. 3
4. 4

Q.33 Read the following features of the Photoelectric effect.
(A) The maximum kinetic energy of the photoelectrons varies linearly with the frequency of incident radiation
(B) The maximum kinetic energy of the photoelectrons is dependent on its intensity.
(C) For a frequency of incident radiation, lower than the cut-off frequency, photoelectric emission is possible even if the intensity is not large.
(D) Ultraviolet light can eject photoelectrons from copper even if visible light does not eject
Choose the correct answer from the options given below:
1. (A), (B) and (C) only
2. (A) and (D) only
3. (A), (B) and (D) only
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.34 The wave theory of light can not explain the phenomenon of
1. refraction of light
2. reflection of light
3. dispersion of light
4. Photoelectric effect

Options 1. 1
2. 2
3. 3
4. 4

Q.35 A moving hydrogen atom absorbs a photon of wavelength 221 nm and stops. If the mass of the hydrogen atom is $1.67 \times 10^{-27}\text{ kg}$,
the linear momentum of the photon will be: (Given: $h = 6.63 \times 10^{-34}\text{ J}\cdot\text{s}$)

1. $3 \times 10^{-27}\text{ N}\cdot\text{s}$
2. $3 \times 10^{-25}\text{ N}\cdot\text{s}$
3. $3 \times 10^{-27}\text{ N}\cdot\text{s}$
4. $3 \times 10^{-26}\text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$

Options 1. 1
2. 2
3. 3
4. 4

Q.36 A solid sphere and a hollow sphere of equal diameters are raised to the same potential, then
1. solid sphere has more charge
2. Both have equal charge
3. hollow sphere has more charge
4. Only hollow sphere has charge

Options 1. 1
2. 2
3. 3
4. 4

Q.37 What will be the expression for the energy of a photon of wavelength λ is:

1. $\frac{hc}{\lambda}$
2. $\frac{\lambda}{hc}$
3. $hc\lambda$
4. $\frac{h\lambda}{c}$

Options 1. 1
2. 2
3. 3
4. 4

Q.38 The angle of minimum deviation if the refracting angle of a prism is 2θ , and the refractive index of the material of the prism is $\cot \theta$ is:

1. $\frac{\pi}{2} - 2\theta$
2. $\pi - 2\theta$
3. $\pi + 2\theta$
4. $\pi - 4\theta$

Options 1. 1
2. 2
3. 3
4. 4

Q.39 Calculate the speed of light in a medium in which the critical angle is 45° :

1. $2.12 \times 10^8 \text{ m/s}$
2. $3 \times 10^8 \text{ m/s}$
3. $3\sqrt{2} \times 10^8 \text{ m/s}$
4. $\sqrt{2} \times 10^8 \text{ m/s}$

Options 1. 1
2. 2
3. 3
4. 4

Q.40 In a cyclotron, if a deuteron can gain an energy of 40 MeV, then a proton can gain an energy of

1. 20 MeV
2. 40 MeV
3. 60 MeV
4. 80 MeV

Options 1. 1
2. 2
3. 3
4. 4

Q.41 Given that two identical coils, one of copper and another of aluminium are rotated with the same angular speed in an external magnetic field, which of the following statements is true?

1. Induced emf will be more in aluminium coil whereas induced current will be more in copper coil.
2. Induced emf will be more in copper coil whereas induced current will be more in aluminium coil.
3. Induced emf and induced current will be same in both the coils.
4. Induced emf will be same in both the coils whereas induced current will be more in copper coil.

Options 1. 1
2. 2
3. 3
4. 4

Q.42 Match List-I with List-II

List-I

List-II

Maxwell's Equations in Vacuum
(Left part)

Maxwell's Equations in Vacuum
(Right part)

(A) $\oint \vec{B} \cdot d\vec{l}$

(I) $-\left(\frac{d\Phi_B}{dt}\right)$

(B) $\oint \vec{E} \cdot d\vec{A}$

(II) 0

(C) $\oint \vec{E} \cdot d\vec{l}$

(III) $\frac{Q}{\epsilon_0}$

(D) $\oint \vec{B} \cdot d\vec{A}$

(IV) $\mu_0 i_c + \mu_0 \epsilon_0 \left(\frac{d\Phi_E}{dt}\right)$

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
2. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1
2. 2
3. 3

Q.43 The speed of the electron in $n = 3$ orbit of He^+ ion would be: (Given $k = 9 \times 10^9 \text{ N}\cdot\text{m}^2\text{C}^{-2}$, $e = 1.6 \times 10^{-19} \text{ C}$, $h = 6.63 \times 10^{-34} \text{ Js}$)

1. $1.46 \times 10^6 \text{ m/s}$
2. $6.41 \times 10^6 \text{ m/s}$
3. $4.46 \times 10^6 \text{ m/s}$
4. $5.23 \times 10^6 \text{ m/s}$

Options 1. 1
2. 2
3. 3
4. 4

Q.44 Out of the following statements, which statements explain the difference between forward biasing and reverse biasing in a p-n junction correctly?

Forward biasing in a p-n junction

Reverse biasing in a p-n junction

(A) p-side is connected to the +ve terminal and n-side to the -ve terminal of the battery.

p-side is connected to the -ve terminal and n-side to the +ve terminal of the battery.

(B) Depletion layer becomes thicker.

Depletion layer becomes thin.

(C) Resistance across the p-n junction increases.

Resistance across the p-n junction decreases.

(D) the current is due to majority charge carriers.

the current is due to minority charge carriers.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (B) and (D) only
3. (A) and (D) only
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.45 The cause of the depletion layer in the p-n junction region is

1. drift of holes
2. drift of electrons
3. migration of impurity ions
4. diffusion of charge carriers

Options 1. 1
2. 2
3. 3
4. 4

Q.46 In the context of the formation of images by spherical mirror, which of the following statements are correct?

- (A) Magnification m is +ve for real images and -ve for virtual images.
- (B) f and R are -ve for a concave mirror and +ve for a convex mirror.
- (C) Magnification m has no units.
- (D) For a real object, u is +ve and v is +ve for real image and -ve for virtual image.

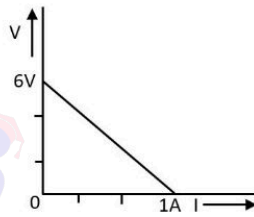
Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (D) only
3. (A), (B), (C) and (D)
4. (B) and (C) only

Options 1. 1
2. 2
3. 3
4. 4

Q.47

Given is the graph of potential difference (V) across combination of three identical cells in series versus current (I). The emf of each cell would be



1. 2 V
2. 3 V
3. 5 V
4. 6 V

Options 1. 1

2. 2
3. 3
4. 4

Q.48 A series LCR circuit with an inductor, a capacitor and a resistor of resistance $50\ \Omega$ is connected to an ac source of 24 V, 50 Hz. The impedance of the circuit is $56\ \Omega$. The average power dissipated in the circuit is

1. 9.18 W
2. 11.5 W
3. 21.4 W
4. 514.3 W

Options 1. 1

2. 2
3. 3
4. 4

Q.49 A radiation of energy E is normally incident on a perfectly reflecting surface.

The momentum transferred to the surface will be: (Given: speed of light c)

1. $\frac{E}{c}$
2. $\frac{2E}{c}$
3. $\frac{E}{c^2}$
4. $\frac{2E}{c^2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.50 The radius of a nucleus varies with its mass number A as:

1. directly proportional to A
2. inversely proportional to A
3. directly proportional to $A^{1/3}$
4. proportional to $\sqrt{A}$

Options 1. 1

2. 2
3. 3
4. 4